Investigating Centripetal Force: Marble and Board Demonstration

Circular motion represents one of the more counterintuitive concepts within the Forces and Dynamics strand of the Grade 12 University Physics (SPH4U) curriculum. Abstract force vectors and mathematical relationships are often difficult for students to visualize until they observe these effects in action. Physical demonstrations help students connect the mathematical principles of circular motion to observable phenomena, revealing how forces interact in rotating systems in a concrete and meaningful way. The motivation for choosing these demonstrations was to connect the physics to real-world experiences that students already recognize. The marble graviton demonstration relates to the amusement park ride "The Gravitron," where riders are pressed against the wall as it spins, showing the same concept that keeps the marble in the glass. The board demonstration also reflects familiar situations, such as swinging a bucket of water at the beach or the motion seen in the Olympic hammer throw, which helps students see how circular motion appears in everyday life.

This P.E.O.E. aligns with expectation **B2.6,** which involves analyzing the forces and acceleration acting on an object in uniform circular motion in both horizontal and vertical planes, and using free-body diagrams and equations to solve related problems. Two demonstrations were used to engage students in these concepts. The first, known as the marble graviton demonstration, involved spinning a wine glass containing a marble so that the marble remained inside even when the glass was inverted. Although simple in appearance, it prompts students to question why the marble does not fall and to reason through the direction and balance of forces acting on it throughout its motion. The second demonstration was inspired by the classic bucket experiment but modified for safety and accessibility. A cardboard board was attached to the string, and small objects were placed on top. Working in groups, students spun the board in a vertical loop and observed that the object stayed on the board instead of falling. Both demonstrations promote critical thinking by requiring students to make and test predictions that often contradict their initial assumptions. Many expect the marble to fall from the inverted glass or believe the object will stay on the board at any speed. When they observe that the marble stays in place and that the object only remains on the board when spun quickly enough, they are encouraged to reconcile these results by analyzing how gravity and the normal force interact at different points in the circular path. This leads to a more meaningful understanding of the relationship between force, acceleration, and velocity, moving beyond memorization toward deeper conceptual reasoning.

In class, this P.E.O.E. would be delivered as a *Minds-On* portion of a lesson that transitions from previous learning about linear motion and forces to the next context of circular motion. Students would already be familiar with free-body diagrams, net force, and acceleration, which allows them to extend their understanding to motion along a curved path. Before the demonstrations, students would receive a brief review of key concepts and information relevant to the activity. The class would then be divided into four groups, with each station equipped with the materials needed to perform the demonstrations. The first activity, the marble graviton demonstration, would begin with a short discussion prompting students to predict what will

happen when a marble inside an inverted wine glass is spun in a circle. Students would record their predictions and explanations, encouraged to make connections to the forces and motion principles studied earlier in the unit. They would then share their reasoning within their groups and then to the class before performing the demonstration themselves. During the activity, I would circulate around the classroom to observe, ask guiding questions, and ensure that students are noticing the intended physical effects. Due to the awkward grip required to continuously spin the glass, the marble can occasionally fly out when the motion is stopped too abruptly. While this still demonstrates the underlying physics, I would prefer that students focus on maintaining steady circular motion to observe the marble staying inside the glass. If a group struggles to achieve this, I would model an alternative approach by placing a hand over the opening of the glass while swirling it, ensuring the marble remains contained while the effect is still visible. After completing the first P.E.O.E., students would discuss and refine their explanations before moving on to the second demonstration involving a cardboard board attached to string with an object placed on top. I would remind students to be aware of their surroundings and to reduce the speed of rotation near the bottom of the circle rather than the top to prevent the object from flying off. Students would repeat the same process of prediction, observation, and explanation as before.

The two demonstrations complement each other, as the marble-in-glass and board-on-string setups illustrate the same underlying principles from different perspectives. In both cases, the motion depends on a continuous inward net force, called the centripetal force, which constantly changes the object's direction without changing its speed. For the marble demonstration, the inward normal force from the glass provides the centripetal force that keeps the marble moving in a circular path, as shown in **Figure 1a**. The marble's inertia causes it to continue in a straight line when the circular motion slows or stops, so if the inward normal force is no longer sufficient to overcome gravity, the marble leaves the glass along a tangential path. In the board-and-string demonstration, shown in **Figures 1b and 1c**, the same physics applies but gravity plays a larger role because the motion occurs in a vertical plane. At the top of the loop, both gravity and the normal or tension force act toward the center to provide the required centripetal force. At the bottom of the loop, gravity acts outward while the normal or tension force acts inward, reducing the net centripetal force. If the object moves too slowly, the inward force cannot maintain the circular path, and the object falls. Comparing these two setups helps students see that centripetal force is not a new or separate force but the resultant of other forces, such as gravity, tension, or the normal force, acting toward the center.

To promote deeper thinking, students could be asked extension questions such as: How does increasing the radius of motion affect the required centripetal force? What would happen to the motion if the speed of rotation were doubled? How do the forces acting on the object differ at the top and bottom of the circular path? What role does mass play in determining the necessary inward force to maintain circular motion? How can these observations be explained using Newton's second law? These questions encourage students to think critically about the

relationships between force, mass, velocity, and radius, helping them build a stronger conceptual understanding of uniform circular motion.

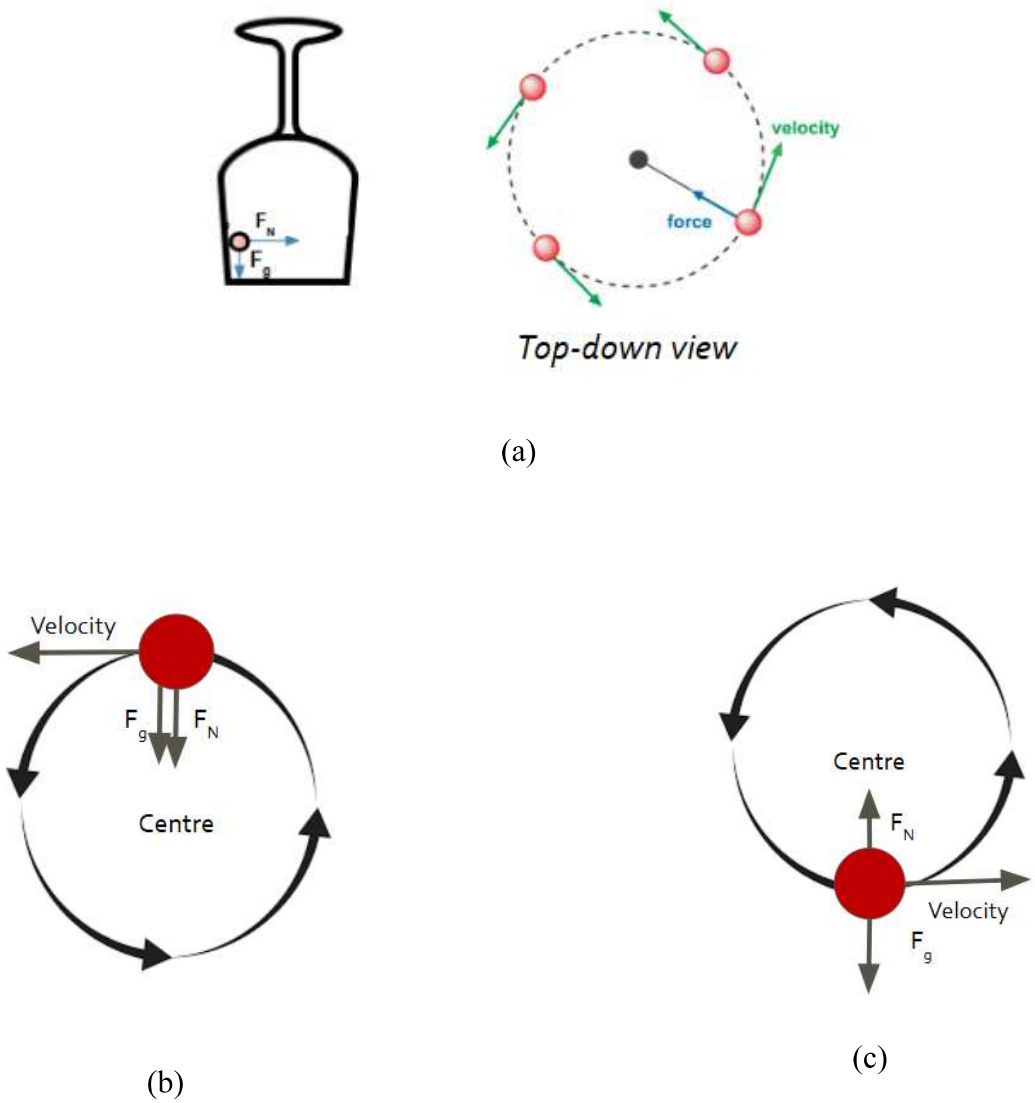

Figure 1. (a) Free-body diagram of the marble-in-glass demonstration showing the marble moving in a horizontal circle. The normal force from the glass acts inward toward the centre of the circular path, providing the centripetal force required to change the marble's direction. The velocity vector is tangent to the circle, and gravity acts vertically downward. (b) Free-body diagram of the board-and-string demonstration at the top of the vertical circle. Both gravity and the normal or tension force act downward toward the center, combining to supply the necessary centripetal force. (c) Free-body diagram of the board-and-string demonstration at the bottom of the vertical circle. Here, gravity acts downward while the normal or tension force acts upward toward the center. The net inward force equals the difference between these two forces, providing the centripetal acceleration needed to maintain circular motion.

Predict	Explain
Observe	Explain